ORIGINAL ARTICLE

TESTING THE CLIMATE VARIABILITY HYPOTHESIS IN THERMAL TOLERANCE LIMITS OF TROPICAL AND TEMPERATE TADPOLES.

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ABSTRACT

Aim The climate variability hypothesis (CVH) states that a positive relationship may exist between the breadth of thermal tolerance range and the level of climatic variability experienced by taxa with increasing latitude, especially in terrestrial ectotherms. Under CVH, we expected to find a correspondence between both thermal tolerance limits (CTmax and CTmin), ambient extreme temperature and the range sizes of species. We examined the validity of these predictions in a lowland tropical and a temperate tadpole assemblage.

Location Lowland Neotropics (Bahia, Brazil) and Palearctic (Iberian Peninsula and North Africa).

Method We employed phylogenetic eigenvector regression (PVR) and Pagel's lambda to analyze phylogenetic signals in CTmax and CTmin. We used phylogenetic regression analyses (PGLS) to test the relationships between thermal limits, range size and temperature predictors (measured at the macroscale and microhabitat levels) and phy-ANOVA to compare both the physiological traits and thermal regimen in both tropical and temperate assemblages.

Results We documented moderate-to-strong phylogenetic signal in both heat and cold tolerance. Temperate-zone tadpoles had broader thermal tolerances than tropical ones. Thermal tolerance range was correlated with range sizes and was explained by seasonal thermal range predictors at the global scale. Both macro- and microclimate temperature variables provided the best predictive multivariate models of thermal limits at the global scale. Microclimatic predictors, however, were the main determinants of CTmax and CTmin variation at the local level of tropical and temperate communities, respectively.

Main conclusions Thermal tolerance range increases with latitude in tadpoles due to the higher increase in cold tolerance in temperate tadpoles. At the global scale, both macro and microenvironment thermal information were reliable predictors of critical thermal limits and thermal tolerance range, as CVH predicts. However, thermal limits were best predicted by temperatures of the micro-habitat at the regional level, thus suggesting that physiological thermal boundaries may be governed by thermal selection.

Keywords CTmax, CTmin, latitudinal variation, macrophysiology, Rapoport's rule, amphibians

INTRODUCTION

Temperature has pervasive effects on life, affecting organisms at different scales ranging from the molecular level, i.e. rates of chemical reaction (Hochachka & Somero, 2002), to the ecosystem level, i.e. ecological interactions (Dunson & Travis, 1991). The study of physiological tolerance ranges, especially thermal ones, is essential to understand many aspects of the biology of organisms. This includes the conditions that define their fundamental niches, geographic distribution and evolutionary dynamics, which in turn allows us to determine their vulnerability to climate change (Hutchinson, 1981; Kearney & Porter, 2009; Soberón & Nakamura, 2009; Peterson, 2011; Duarte *et al*., 2012; Seebacher & Franklin, 2012).

The interest in the study of the evolution and functionality of thermal limits has led to the development of numerous biogeographical hypotheses (Gaston *et al*., 2009). In a seminal paper, Janzen (1967) suggested that mountains would act as physiological barriers to the dispersal of tropical species. This prediction originally derived from the observation that annual climatic variation is relatively lower in the tropics than at higher latitudes, so that tropical organisms should have narrower physiological thermal breadths compared to organisms in temperate zones. This idea was later on generalized with the climate variability hypothesis (CVH) proposed by Stevens as a potential explanation for Rapoport's rule (Stevens, 1989). Climate variability hypothesis predicts that temperate species would be able to inhabit larger geographic ranges than tropical ones (Stevens, 1989) suggesting a positive relationship between thermal tolerance range and the level of climatic variability experienced by taxa with increasing latitude (Bozinovic *et al*., 2011).

78 The CVH hypothesis has been tested for insects (Addo-Bediako *et al*., 2000;
79 Calosi *et al*., 2010; Sheldon & Tewksbury, 2014), bivalves (Compton *et al*., 2007),
80 lizards (Van Berkum, 1986; Cruz *et al*., 2005; Clusella-Trullas *et al*., 2011) and
81 amphibians during their adult terrestrial stage (Snyder & Weathers, 1975; Ghalambor *et*
82 *al.,* 2006, both reanalyzing the original data set by Brattstrom, 1968; John-Alder *et al.,*
83 1988). Two recent global-scope publications have synthesized our knowledge on this
84 topic: Sunday *et al*. (2014) for aquatic and terrestrial ectothermic animals and Araújo *et*
85 *al*. (2013) for terrestrial ectotherms, including plants. Most of these studies suggest that
86 increasing thermal tolerance in extratropical ectotherms basically depends on the
87 extreme reduction in CTmin with latitude, attributed to a rapid decline in minimum
88 ambient temperatures. Contrarily, the variation in maximum ambient temperature seems
89 to be relatively independent of latitude, making CTmax geographically invariant
90 (Gaston *et al*., 2009). This asymmetric pattern appears to be a general trend (Addo-
91 Bediako *et al*., 2000; Huey *et al*., 2009; Clusella-Trullas *et al*., 2011; Araújo *et al*.,
92 2013) and has even been considered a rule in terrestrial ectotherms (Brett's rule, Gaston
93 *et al*., 2009). However, quite interestingly, this trend is not supported by aquatic marine
94 organisms which exhibit similar reductions in both thermal limits with latitude (Sunday
95 *et al*., 2011). This asymmetry may have important biogeographical implications. For
96 instance, the ability of species to evolve physiological tolerance to cold may condition
97 the migration of lowland tropical lineages to temperate zones or highland tropical
98 climates (e.g Wiens *et al*., 2006).

99 In this study we test the CVH prediction of an asymmetrical latitudinal variation
100 for thermal tolerance limits and explore their relationship with climatic determinants in
101 two larval amphibian assemblages from different climatic regimes: a temperate and a
102 lowland tropical assemblage. Aquatic larval amphibians are an ideal model to analyze

103 thermal adaptations. Despite being capable of regulating their body temperatures
104 (Hutchison & Dupré, 1992), tadpoles may be limited in their search for favorable
105 microhabitats, for instances when trapped in shallow heated ponds where tadpoles are
106 obligated to act as thermoconformers (Balogová & Gvoždík, 2015). Thus, their
107 physiological resistances may have been adjusted to the local thermal extremes
108 experienced in their ponds through thermal selection.

109 It is expected that local thermal conditions drive the evolution of thermal
110 tolerance limits which ultimately result in thermal adaptations (Angilletta, 2009;
111 Bozinovic *et al*., 2011). Assuming that natural selection governs the evolution of
112 thermal resistances, an asymmetric geographical variation in thermal tolerance limits
113 will emerge if two predictions are met. First, there exists a correspondence between
114 either thermal tolerance limits with ambient extreme temperature and, second, that such
115 correspondence is steeper for cold tolerance and weaker for warming tolerance
116 (Kellermann *et al*., 2012b; Araújo *et al*., 2013; Sunday *et al*., 2014). One way to
117 determine whether thermal selection is prone to drive thermal tolerance limits is by
118 assessing the risk of populations and species to suffer heat or cold impacts. An operative
119 metric to estimate the eventual occurrence of heat shock is warming tolerance (i.e. the
120 difference between CTmax and Tmax) (Deutsch *et al*., 2008; Duarte *et al*., 2012).
121 Similarly, we can define cooling tolerance as the risk to suffer cold shocks and measure
122 it as the difference between the minimum recorded temperature, Tmin, and the CTmin.

123 Previous approaches dealing with CVH predictions may be weakened by some
124 concerns. These rely on meta-analyses employing heterogeneous and non-standardized
125 procedures that may yield biased estimates of thermal tolerances (Rezende *et al*., 2011),
126 and also depend on limited phylogenetic controls. Additionally, most approaches
127 dealing with the relationship between thermal resistance traits and environmental

128 variables rely on macroclimatic information from large-scale databases or repositories
129 (e.g. WorldClim) (Clusella-Trullas *et al*., 2011; Kellermann *et al*., 2012a, 2012b;
130 Araújo *et al*., 2013). These may not fairly reflect the operative temperatures to which
131 ectotherms are exposed, especially when dealing with extreme temperatures where
132 thermoregulation may play a significant role (Sunday *et al*., 2014). Therefore,
133 macroclimate estimators should provide lower explanatory power than microclimate
134 data, especially when analyzing thermotolerance on an individually-based local area
135 (Angilletta *et al*., 2002; Graae *et al*., 2012; Navas *et al*., 2013; Potter *et al*., 2013). In
136 order to better characterize environmental thermal conditions, we gathered temperature
137 data from both macroclimate (WorldClim) and microclimate (water temperature
138 measurements) during the time tadpoles are present in the ponds.

139 A key factor to determine species vulnerability to climate change is their
140 capacity to adapt their thermal tolerance limits through plastic and/or evolutionary
141 responses (Williams *et al*., 2008; Hoffmann *et al*., 2012). As a result of climate niche
142 conservatism, however, closely related species may display similar physiological
143 resemblance and limited adaptive potential (Losos, 2008). A way to test the
144 evolutionary constraints of physiological traits is the study of phylogenetic signal
145 through comparative methods. For example, CTmin has been considered a more labile
146 evolutionary trait than CTmax in ectotherms (Araújo *et al*., 2013; Grigg & Buckley,
147 2013). Here we use a statistically-based phylogenetic eigenvector regression method
148 (Diniz-Filho *et al*., 1998) to infer the relative weight of heredity versus adaptation on
149 thermal limits.

150 We aim to answer the following related questions and predictions:

i) Did temperate tadpoles evolve wider thermal and geographical ranges through increasing cold tolerance as expected from CVH (Janzen, 1967) and Rapoport's rule (Stevens, 1989)? ii) What set of thermal predictors (macro or micro) have a higher explanatory power to describe the variation in thermal limits and thermal range of species at both large latitudinal and regional ranges? iii) Given the latitudinal variation of seasonality, should we expect stronger correlations between lower thermal limits and minimum environmental temperature than for upper thermal resistance and maximum temperature?

METHODS

Amphibian tadpole surveys and estimates of thermal tolerance

Thermal tolerance was determined for tadpoles of 47 species. We examined 27 tropical species in the state of Bahía (Brazil) between latitudes 13º-15º S from November 2011 to January 2012. The studied species occupy different aquatic environments with contrasting temperatures and thermal variability (see Table S2, Appendix S2 in Supporting Information) such as forest ponds and streams in the Atlantic Forest (Mata Atlântica) and open forest ponds both in the semi-arid Caatinga (Contendas do Sincorá FLONA reserve), and in deforested areas of Mata Atlântica. We also included two species inhabiting phytotelmata associated to open forest Atlantic coast restingas (Table S2 in Appendix S2). We studied 20 temperate species in the Iberian Peninsula and northern Morocco ranging from latitude 29º-43º N, from 2011-2013 (Table S2, Appendix S2). These temperate species show different breeding phenologies and also inhabit a variety of thermal environments from mountain ponds, streams, Mediterranean lowland ponds to semi-arid sub-Saharan species (Table S2, Appendix S2). All examined tadpoles were collected in their natural ponds and transported to one of the two reference laboratories at particular study sites (Brazil, UESC, Ilhéus, November 2011 to January 2012; Spain, EBD-CSIC, Sevilla, 2011-2013). During all the experiments, larvae were maintained at a similar and constant room temperature of 20 °C, with a natural photoperiod ca. 12:12 light: dark. Tadpoles were placed in plastic containers with similar densities, for a minimum of three-four days. This acclimation period was chosen as previous research in adult amphibians revealed that between 2-3 days was the time required to stabilize CTmax and CTmin after a large change in acclimation temperature, such as field and laboratory environments (Brattstrom, 1968). Larvae were tested individually between 25 and 38 Gosner stages (Gosner, 1960). Tadpoles over 38

185 Gosner stage were excluded because near the metamorphic climax they tend to have
186 reduced thermal tolerances (Floyd, 1983).

187 Whenever possible, we used a minimum sample of 16 individual tadpole
188 replicates per species and thermal tolerance limit (although for some species, mostly
189 tropical, the available number of individuals examined was lower, Table S1, Appendix
190 S2). Each tadpole was weighed immediately before the beginning of the test. Both
191 thermal tolerance limits were determined using the Hutchison's dynamic method
192 (Lutterschmidt & Hutchison, 1997a) in which each animal was exposed to a constant
193 heating / cooling rate (ΔT=0.25 °C min^{-1}) until an end point is attained. The end-point
194 was signalled for both thermal limits as the point at which the tadpoles become
195 motionless and fail to respond to external stimuli by prodding 10 consecutive hits
196 applied each two seconds with a wooden stick. Each tested tadpole was placed
197 individually in 100 mL containers with dechlorinated tap water in a refrigerated heating
198 bath of 15 L (HUBER K15-cc-NR) at a start temperature of 20 °C (temperature of
199 acclimation). Because of the small size of tadpoles we assumed that body temperature
200 was equivalent to water temperature (Lutterschmidt & Hutchison, 1997b) and then
201 CTmax and CTmin were recorded as the water temperature beside the tadpole measured
202 with a Miller & Weber quick-recording thermometer (to the 0.1 °C). After a tolerance
203 limit was determined, we immediately transferred tadpoles to water at the acclimation
204 temperature (20 °C) to allow for recovery, after which their Gosner stage was
205 registered. Tadpole survival was verified a few minutes and 24 hours after the end of the
206 heating/cooling assays. Each individual was tested only once. To ensure that lethal
207 temperature was not exceeded only those individuals who recovered 24 h after the test
208 were included in subsequent analyses. Although we only examined a single population
209 for each species, we assumed that response variation among species is larger than

variation within species (cf. Klok & Chown, 2003). In a large multi-species comparison, any intraspecific difference in CTmax and CTmin would be overwhelmed by the high number of species and the great diversity of thermal environments examined. To analytically address this potential issue, we explored inter-populational variation for those species in our dataset for which we had available data. Analyses on different populations of both temperate and tropical species (*Rana temporaria*, *Bufo bufo*, *Epidalea calamita*, *Pelodytes ibericus*, *Rana arvalis* and *Dendropsophus novaisi*) revealed that intraspecific variability in thermal limits is rather limited (around 1 °C) compared to the variation observed among species from the same region (5.9 °C and 5.3 °C for CTmax in temperate and tropical communities, respectively, and 4.7 °C and 5.7 °C for CTmin in temperate and tropical communities respectively) or differences between temperate-tropical regions (8.3 ºC for CTmax and 11.8 ºC for CTmin). Finally, we calculate the thermal tolerance range as the difference between CTmax and CTmin for each species.

Climatic determinants of thermal tolerance limits

To obtain macroclimatic determinants of thermal tolerance limits we used digital distribution maps from IUCN (IUCN and Nature Serve, 2006) and processed them in QGIS to obtain the following geographical variables for each species: geographic range size, longitude and latitude of the centroid, maximum and minimum values of latitude, and latitudinal range. Latitudinal and longitudinal coordinates of population samples were taken to extract and summarize the climatic information from WorldClim layers (30 ” spatial resolutions; records from 1950 to 2000 (Hijmans *et al*., 2005)). Although WorldClim climatic variables correspond to air measurements, previous studies have shown that air temperatures correlate well with water temperature in streams and lakes (Livingstone & Lotter, 1998; Webb *et al*., 2003), being reliable predictors used in

biogeographical and conservation analyses with aquatic organisms such as continental fishes (Chessman, 2013) and tadpoles (Gerick *et al*., 2014). We selected the following four macroclimatic variables: TMAX (Maximum of the Average monthly maximum temperature); TMIN (Minimum of the Average monthly minimum temperature); TMEAN (Monthly Mean Temperature) and Temperature SEASONAL RANGE (TMAX-TMIN). For each species only the months in which tadpoles were present in the ponds were taken into account to perform the analyses (for further details see Appendices S1 and S2).

To assess the explanatory power of microclimatic habitat conditions on thermal tolerance limits, we included temperature data obtained for each sample points directly from dataloggers (HOBO pendant) that were deployed at the bottom of diverse aquatic microenvironments such as ponds, streams and phytotelmata. Temperature was recorded every 15 min. We analyzed mean (tmean), maximum (tmax) and minimum (tmin) daily temperatures, average daily range and seasonal range (tmax-tmin) from each water body for the period in which tadpoles were present in the aquatic habitat. The number of sampling days ranged from 30-497 days (see Table S2 Appendix S2). To evaluate the risk of each species suffering cold and heat acute shocks, we estimated both warming and cooling tolerances (sensu Deutsch *et al*., 2008; Duarte *et al*., 2012) as (CTmax - tmax) and (tmin - CTmin), respectively.

Statistical analyses

Phylogeny

We constructed a phylogenetic tree containing the 47 studied species from Pyron & Wiens (2011). For 16 species not included in this phylogeny, we used the position of a known sister-taxon that does appear in the phylogeny. In some cases the position of

259 certain species within the genus could not be resolved and therefore appear as
260 polytomies (Fig. S1, Appendix S3).

261 *Phylogenetic signal on thermal limits*

262 We tested for phylogenetic signal in each trait calculating Pagel's λ (with the
263 fitContinuous function in the R package GEIGER (Harmon et al., 2008)), which assumes
264 a Brownian motion model of character evolution. Apart from this model-based metric,
265 we also incorporated a statistical-based phylogenetic method that makes no assumptions
266 on the underlying evolutionary model to evaluate the weight of heredity versus
267 adaptation in the critical thermal limits in our data. Phylogenetic eigenvector Regression
268 (PVR) (Diniz-Filho *et al*., 1998), allowed us to partition the components of variance
269 attributable to ecological (S) and phylogenetic effects (P). We used the function
270 PVRdecomp from the PVR package (Santos *et al*., 2013) in R (see Appendix S1).

271 *Phylogenetic comparative analyses*

272 To evaluate the correlations between thermal physiology variables (CTmax, CTmin and
273 thermal tolerance range), and geographical variables (area and latitudinal range of
274 species distribution, latitude of the centroid distribution and maximum latitude), we
275 used phylogenetic generalized least squares (PGLS) analyses under a Brownian motion
276 model of evolution using the R package CAIC (Orme *et al*., 2009). We also used PGLS
277 to test for correlation between CTmax and CTmin. A multiple regression approach by
278 PGLS, was used to examine relationships between the physiology variables and the
279 environmental temperature variables (both for WorldClim and dataloggers). We
280 selected the best model using the lowest Akaike information criterion (AIC) (Burnham
281 & Anderson, 2002). See Table S7-S9, Appendix S2 for a summary of the models used.

To analyze the influence of climatic region (tropical versus temperate) on the variation of both physiological traits (CTmax, CTmin, Thermal tolerance range, and Warming and Cooling Tolerances) and environmental temperature variables, we used phylogenetic ANOVA models (phy-ANOVA) in CAIC. Previous analyses revealed that tadpole mass did not explain significant CTmax nor CTmin variation and, hence, was not included in the models. We employed ANCOVA to compare differences in the regression slopes of CTmax / CTmin with latitude using the basic R package (see Appendix S1).

RESULTS

Climate Variability and Rapoport's Hypothesis in tadpoles: comparison of thermal tolerance between temperate and tropical geographical areas

The two methods we used to estimate phylogenetic signal (Pagel's lambda and PVR) identified strong phylogenetic effects on CTmin (λ = 0.97, P component = 0.80), whereas CTmax showed only moderate levels of phylogenetic signal (λ = 0.67, P = 0.50) (Table S5, Appendix S2). Therefore, PGLS was used in all subsequent analyses to control for the phylogenetic structure in the data.

Temperate species showed on average greater thermal tolerance ranges than tropical species (phy-ANOVA, P <0.01) (Fig. 1 and Table 1). Broader thermal breadths in temperate species were mainly attributed to their higher resistance to cold (phy-ANOVA, P <0.001, Table 1), which offsets their lower values for CTmax (phy-ANOVA, P <0.01, Table 1). Both CTmax and CTmin decreased with latitude (Fig 2a, Table S4 Appendix S2), but the slope of the variation in CTmin (-0.20 ± 0.02) was significantly steeper than that of CTmax (-0.08 ± 0.02) when both are compared against species' latitudinal centroids (ANCOVA: CT [min/max] x Latitude, $F_{1,86}$ = 7.38, P

306 <0.01). However, neither CTmax nor CTmin were correlated with latitude when each
307 geographical area was analyzed separately.

308 Analyses performed with combined data (i.e. tropical plus temperate) showed
309 that thermal tolerance range was correlated with species maximum latitude, latitudinal
310 range and range size (log transformed) (PGLS, $F_{1,36}$ = 43.85, P <0.01; $F_{1,36}$ = 123.4, P
311 <0.01, $F_{1,36}$ = 148.8, P <0.01, respectively) (Table S4, Appendix S2, Fig. 2b and Figure
312 3). However, when analyses were restricted to a particular area, only the temperate
313 assemblage marginally exhibited the trend with latitudinal range (PGLS, $F_{1,15}$ = 4.19, P
314 = 0.06).

315 **Macro and microclimatic predictors for the thermal tolerance traits and estimates**
316 **of warming and cooling tolerances**

317 Maximum ambient temperatures did not differ between geographical areas (in both
318 macroclimatic and microclimatic approaches) (Table 1), (see also Fig. S3a, Appendix
319 S3). Mean and minimum temperature (for both macro and microenvironmental scales)
320 were significantly lower for the temperate assemblage (phy-ANOVA, P <0.001, Table
321 1, Fig. S3b, Appendix S3). Although average daily range did not differ between regions
322 (Table 1), both SEASONAL RANGE (from macroclimate) and seasonal range
323 (microclimate) was significantly higher for temperate latitudes (phy-ANOVA, P <0.01)
324 (Table 1). Multivariate models that combined macro- and microclimatic temperature
325 variables, showed the best predictive power for physiological traits at the extent of
326 whole latitudinal range (Table 2). CTmax was best predicted with TMEAN and tmax
327 (CTmax ~ TMEAN + tmax, AIC = 113.1, $F_{2.35}$ = 18.56, P <0.01, Table 2). The best
328 model for CTmin included minimum temperature from macro- and microclimate
329 (CTmin ~ TMIN + tmin, AIC = 128.2, $F_{2.35}$ = 153.80, P <0.01, Table 2). Thermal

tolerance range was best predicted by seasonal range (macro and microclimate) (Tolerance Range ~ SEASONAL RANGE + seasonal range, AIC = 158.05, $F_{2,35}$ = 17.85, P <0.01, Table 2). By examining the slopes of climatic predictors of tolerance limits for pooled communities, we found steeper trends for CTmin predictors than for CTmax (Table 2).

Within-region analyses revealed that thermal limits correlated better with micro- than macroclimatic temperature predictors for the most extreme temperature in each zone: tmax with CTmax (tropical assemblage) and tmin with CTmin (temperate assemblage). Thermal tolerance range was not correlated with either micro- or macroclimatic ambient temperature at local scales (Table 2). Similar to the pooled analyses, the slopes of climatic predictors within particular communities were higher for CTmin than for CTmax (Tables 2, Fig. S3, Appendix S3). The analyses of warming tolerance between communities revealed that tropical and temperate species did not differ (phy-ANOVA, $F_{1,42}$ = 0.185, P = 0.89, Table 1). However, by distinguishing open and forest pond microenvironments within the tropical assemblage, we found that open forest pond species had lower warming tolerances than both temperate species and tropical forest pond species (phy-ANOVA, $F_{2,41}$ = 28.253, P = 0.01, Table S6, Appendix S2, Fig. S4, Appendix S3). Cooling tolerance was much lower for temperate assemblage species than tropical ones at both microenvironments (phy-ANOVA $F_{2,41}$= 192.53, P < 0.0001; Table S6, Appendix S2, Fig S4, Appendix S3).

DISCUSSION

Thermal tolerance range and the distribution of species: CVH as an explanation for Rapoport's rule

353 Our overall analyses revealed that temperate species have greater thermal tolerance
354 ranges than tropical ones, thus supporting the climatic variability hypothesis (Janzen,
355 1967; Stevens, 1989). The observed increase in thermal tolerance range for temperate
356 species compared to tropical ones stems from their greater cold resistance, rather than
357 an equivalent increase in upper tolerances. This asymmetric latitudinal trend in lower
358 and upper thermal resistance limits is concordant with previous analyses for amphibians
359 during their adult terrestrial stage (Snyder & Weathers, 1975; Ghalambor *et al*., 2006),
360 who reanalysed Brattstrom's (1968) dataset, thus suggesting a similar trend for both
361 aquatic and terrestrial amphibian life stages. Similarly, other terrestrial ectotherms
362 exhibit clear latitudinal clines in CTmin, whereas weaker or even no latitudinal
363 variation for CTmax (Addo-Bediako *et al.*, 2000; Huey *et al*., 2009; Clusella-Trullas *et*
364 *al.*, 2011; Sunday *et al.*, 2011; Araújo *et al*., 2013). This pattern is considered a
365 macrophysiological rule in terrestrial ectotherms although it was originally formulated
366 in fishes (Brett, 1956; Gaston *et al*., 2009) and it is argued that the increased thermal
367 tolerance range in temperate species mainly results from higher temperature
368 seasonalities as they move away from the equator. Such seasonality emerges due to the
369 rapid drop of minimum temperatures whereas maxima are relatively independent of
370 latitude (Chown *et al*., 2004; Olalla-Tárraga *et al*., 2011; Whitton *et al*., 2012).

371 Thermal tolerance range was predicted by thermal variability from both macro-
372 and microclimate estimators, which is consistent with CVH predictions. Although
373 tropical and temperate regions showed a similar daily thermal range at the micro scale,
374 seasonal range (both at macro- and microclimatic levels) was higher for the temperate
375 area. Interestingly, the level of daily thermal variation at the micro scale for the tropical
376 assemblage was twice than that found in the temperate one (CV 108.9 % and 40. 9%,
377 respectively). This reflects the high diversity of breeding habitats of tropical amphibians

(Scheffers *et al*., 2013). Within the studied lowland tropical area (where minimum temperatures did not drop below 17ºC), we found habitats of high thermal stability (e.g. ponds and streams located under the forest canopy of Mata Atlântica) with mean daily ranges extending from 0.1 ºC – 1.6 ºC, well below the average range of the temperate zone (4.7 ºC). However, other biomes and microenvironments studied in the tropical region, such as open ponds in Mata Atlântica, Caatinga, and the specialized phytotelmata showed similar or even greater temperature daily ranges to that found in the temperate assemblage (they can reach up to 41.3 ºC, with a daily oscillation of 7.3-10.3 ºC) (Table S2, Appendix S2). In contrast, in the temperate zone, seasonal range was greater due to the drop of minimum temperatures during autumn-winter in contrast to the higher temperatures reached in spring and early summer. Thus, increased thermal variability in the temperate zone was mainly due to the seasonal component associated to minimum temperature decreases in winter (to 0.4 ºC).

Our overall results controlling for the phylogenetic structure in the data show that thermal tolerance range may explain a significant variability of both range size and latitudinal extent, especially for temperate species. This supports CVH as an explanation for Rapoport's rule. Similar results have been found in adult aquatic insects (Calosi *et al*., 2008, 2010) and terrestrial ectotherms such as *Liolaemus* lizards (Cruz *et al*., 2005).

Average CTmax for temperate species was lower than expected by maximum ambient temperatures, which showed no significant differences between tropical and temperate areas. Interestingly, tropical and temperate species exhibit a parallel trend of CTmax with tmax but tropical species have relative higher tolerances (conventional ANOVA, $F_{1,41}$ = 70.28, P <0.01; LS means (mean ± SE, N), Tropical: 40.99ºC ± 0.22ºC, N=27; Temperate: 38.37ºC ± 0.28ºC, N=17, Fig S3, Appendix S3). Two complementary arguments may account for the pattern of lower CTmax found for

403 temperate species. The first argument is ecological: while maximum temperature values
404 did not differ between both environments, the timing of ocurrence of heat peaks over
405 the larval season shows differences. High temperatures in the ponds can be reached at
406 any time in the tropics, but the occurrence of maximum temperatures is nonetheless
407 predictable and mainly restricted to late spring and early summer at temperate latitudes.
408 In addition, warming tolerances of the tropical open forest species lay near the lethal
409 threshold in which physiological thermal mechanisms could hardly overcome high
410 temperatures (Fig. S3, S4, Appendix S3). Thus, selection on upper thermal limits is
411 probably more intense for tropical tadpole species, whereas temperate amphibians can
412 temporarily tune their larval period and shift their breeding time to avoid risky heating
413 peaks, which may explain their greater safety margins. Thereby, we found better fitting
414 between CTmax and tmax for the tropical species (Table 2 and Fig. S3a, Appendix S3).
415 The second explanation may rely on the existence of a physiological trade-off between
416 thermal limits, so that achieving greater cold tolerance has an associated cost in
417 tolerance to high temperatures, as suggested by the positive correlation found between
418 CTmax and CTmin in the pooled sample. This trade-off has been explained by
419 molecular mechanisms such as changes in the structure of enzymes and other proteins,
420 when adjusted to low or high temperatures; the modification of fluidity and viscosity of
421 the cell membrane through its lipid composition; or oxygen limitation (Angilletta,
422 2009). Terrestrial and aquatic ectotherms seem to show opposite conclusions regarding
423 this trade-off (Hoffmann *et al*., 2012). While marine species tend to have coupled
424 physiological limits (Portner *et al*., 2006; Sunday *et al*., 2011), such correlation has not
425 been detected in terrestrial species (mainly insects) (Alford *et al*., 2012), but see
426 Anderson *et al.* (2003) for the molecular basis of a trade-off in thermal tolerance in
427 *Drosophila* and an example of evolutionary thermal specialization in *Escherichia coli*

(Bennett & Lenski, 1993). Our results are in consonance with those obtained for other aquatic organisms.

Macroclimatic and microclimatic predictors of thermal tolerance

Models combining climatic predictors at both macro and microclimate scales provided the best explanatory power of physiological thermal limits for global comparisons. Interestingly, analyses restricted to a single region concluded that microclimate variables were more predictive for thermal limits (CTmax in the tropical assemblage and CTmin in the temperate assemblage), which are presumably under high thermal selection, as the lower values of warming and cooling tolerance may suggest. This implies that the evolution of heat tolerance for tropical and especially cold resistance for temperate tadpoles may be governed by thermal extremes experienced in the pond microenvironments.

Given the closer relationship of latitude with lower tolerance, we predicted a stronger correlation between environmental predictors and CTmin than with CTmax (Araújo *et al*., 2013). Accordingly, we found weaker correlation coefficients for CTmax and temperature variables predictors. As we expected, the best predictor for CTmin was minimum temperature both from micro- and macroclimatic data. Our results agree with those previously found for adult amphibians, reptiles, and *Drosophila,* where CTmin was related to minimum or mean temperatures (Clusella-Trullas *et al*., 2011; Kellermann *et al*., 2012a; Araújo *et al*., 2013). CTmax was correlated, in our study, with both maximum temperatures and mean temperature at the micro- and macroclimatic scale. By contrast, in reptiles, CTmax was more related to predictors of temperature variation such as seasonality or diurnal temperature range (Clusella-Trullas *et al*., 2011). In adult amphibians, Araújo *et al*. (2013) found a positive correlation between CTmax

and Tmax whereas the slope was nearly zero for Tmean. In *Drosophila*, Kellermann *et al*. (2012b) found an interaction between maximum temperatures and precipitation as the best predictor. Therefore, this diversity in climatic predictors of upper thermal resistance limits in different ectotherms may probably be conditioned by the specific physical environment (terrestrial or aerial) to which they are exposed, which also limits their ability to use behavioural compensation (the Bogert effect, Huey *et al*. 2012; Sunday *et al*, 2014), so that they cannot be generalized for all ectotherms.

Phylogenetic considerations

Recent studies have found strong phylogenetic signal in the realized climatic niches of amphibians worldwide (Hof *et al*., 2010; Olalla-Tárraga *et al*., 2011; Gouveia *et al*., 2014). Using taxonomic and phylogenetic analyses, Olalla-Tarraga *et al*. (2011) found that the climatic niche dimension that tends to be most phylogenetically conserved is cold tolerance. Only a few exceptions within each taxon (termed as 'escapee' lineages) have apparently been able to shift their cold tolerance and occupy northern latitudes. Strong phylogenetic conservatism in cold tolerance has previously been reported for hylid frogs as well (Smith *et al*., 2005; Wiens *et al*., 2006). Our results on the fundamental thermal niche for tropical and temperate amphibian tadpoles are in agreement with these findings and showed moderate phylogenetic signal for CTmax and strong signal for CTmin. Contrastingly, previous work on lizards has shown lower values of phylogenetical signal for CTmin (Grigg & Buckley, 2013), although Kellermann *et al*. (2012a) also found a moderate signal for *Drosophila*. Overall, it is often assumed that CTmax is an evolutionary conservative trait (Hoffmann *et al*., 2012), whereas CTmin tends to be more plastic.

Concluding remarks

476 Previous works have analyzed how macroclimatic conditions determine thermal
477 physiology traits. However, most organisms, especially small ectotherms with limited
478 dispersal ability, do not experience climate on geographical scales much larger than
479 organismal size. Rather, they live under microclimatic conditions, which can be highly
480 heterogeneous and differ from surrounding macroclimates (Potter *et al*., 2013). In this
481 study both macro- and microclimate temperature variables were reliable predictors of
482 thermal limits at the global scale and, as CVH's hypothesis predicts, thermal range scale
483 with species geographical range. However, physiological limits variation at specific
484 regional community is much better explained by microclimate variables related to the
485 prevalent local most stressful thermal conditions (heat in the tropical and cold in the
486 temperate region, respectively) which suggest both the mismatch between organismal
487 scale and climatic data and that physiological thermal boundaries may be governed by
488 thermal selection operating at the local environment. Further studies are required to
489 establish how the tolerance–habitat condition–macroclimate relationships are
490 interrelated and scales up from physiological parameters to the geographical distribution
491 of the species.

492 Under current scenarios of climatic change it is expected greater increases in
493 mean environmental temperatures and the frequency of extreme thermal events in the
494 future (Pachauri *et al*., 2014). Given the conservatism of thermal limits, especially
495 upper tolerance (Hoffmann *et al.*, 2012), and the low overall plasticity potential of
496 ectotherms (Gunderson & Stillman, 2015), it seems unlikely that amphibians may
497 mitigate climatic changes relying solely on thermal adaptation of critical thermal limits.
498 Upward and poleward migration, tracking suitable temperatures may be a chance for
499 certain ectotherms (Parmesan, 2006), but probably not for organisms with poor
500 dispersal ability such as amphibians (Sinsch, 1991). This limitation is evidenced, for

example, by the fact that current distributions of amphibians display high levels of non-equilibrium with current climate (Araújo & Pearson, 2005). New thermal conditions created by climatic change surely will limit southern boundaries of taxa's latitudinal range or even threat population with local extinctions, especially lowland tropical species, as for example those inhabiting the tropical open forest Caatinga. However it could eventually allow the poleward expansion of other species currently limited by low temperatures with the existence of suitable pathways for dispersal (Araújo *et al*., 2006).

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751

752 **SUPPORTING INFORMATION**

753 Additional Supporting Information may be found in the online version of this article:

754 Appendix S1 Supplementary Methods.

755 Appendix S2 Supplementary Tables.

756 Appendix S3 Supplementary Figures.

757 **BIOSKETCH**

758 **Luis M. Gutiérrez-Pesquera** is a PhD student in evolutionary ecology department at
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763 Author contributions: L.M.G.P., M.T. conceived the idea. L.M.G.P., M.T., H.D., M.S.,
764 A.G.N. provided physiological and microclimatic data. L.M.G.P., M.T., M.A.O.T.
765 conducted statistical analyses. L.M.G.P. led the writing of the manuscript with critical
766 participation of all co-authors.
767 **Editor**: Richard Ladle
768
769

770 **TABLES**

771 **Table 1.** Phylogenetic ANOVAs comparing physiological traits (CTmax, CTmin,
772 tolerance range, warming and cooling tolerance) and climatic predictors (macroclimate
773 WorldClim estimators in caps and microenvironment predictors from dataloggers, in
774 lowercase), considering only the reproductive period where tadpoles are present,
775 between tropical and temperate tadpole assemblages. TMAX (maximum of the average
776 monthly maximum temperature); TMIN (minimum of the average monthly minimum
777 temperature); TMEAN (mean of the average monthly temperature); SEASONAL
778 RANGE (TMAX-TMIN); tmax (maximum temperature); tmin (minimum temperature);
779 tmean (mean temperature); daily seasonal range (mean seasonal daily tmax-tmin);
780 seasonal range (seasonal absolute tmax - absolute tmin).

781

	Temperate			Tropical				
Physiological traits	N	$\bar{X}$	SD	N	$\bar{X}$	SD	F-value	Phy-p
1. CTmax	20	38.5	1.46	27	41.10	1.37	39.371	0.01
2. CTmin	20	0.50	1.43	27	6.90	1.29	260.51	0.001
3. Tolerance Range	20	38.0	1.57	27	34.2	1.89	54.028	0.004
4. Warming Tolerance (WT)	17	9.24	1.41	27	10.06	1.22	0.185	0.89
5. Cooling Tolerance (CT)	17	3.10	0.31	27	13.15	0.35	389.72	0.001
Macroclimatic temperatures (WorldClim)								
1.TMAX	20	25.12	5.345	27	29.51	0.45	18.277	0.11
2.TMIN	20	3.99	3.39	27	18.71	1.19	435.75	0.001
3.TMEAN	20	12.83	2.29	27	24.3	0.5	645.10	0.001
4. SEASONAL RANGE	20	21.1	6.03	27	10.8	1.5	73.64	0.001
Microclimatic temperatures (Dataloggers)								
1.tmax	17	30.04	6.27	27	31.03	7.18	0.22	0.891
2.tmin	17	3.35	2.17	27	20.03	1.29	1024.7	0.001
3.tmean	17	12.52	2.57	27	24.07	1.91	291.1	0.001
4.daily seasonal range	17	4.74	1.94	27	2.48	2.70	8.9092	0.315
5.seasonal range	17	26.69	6.12	27	11.00	7.45	52.801	0.01

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783 **Table 2.** Multiple regression models (PGLS) used to predict physiological variables (CTmax, CTmin and tolerance range) with environmental
784 thermal information at different spatial scales in tadpoles. For each variable in the model we include their standardized coefficients to evaluate
785 the relative importance of each one. Climatic predictors (macroclimate WorldClim estimators in caps and microenvironment predictors from
786 dataloggers, in lowercase), considering only the period where tadpoles were present. TMAX (maximum of the average monthly maximum
787 temperature); TMIN (minimum of the average monthly minimum temperature); TMEAN (mean of the average monthly temperature);
788 SEASONAL RANGE (TMAX-TMIN); tmax (maximum temperature); tmin (minimum temperature); tmean (mean temperature); daily range
789 (mean seasonal daily tmax-tmin); seasonal range (seasonal absolute tmax - absolute tmin). λ = Pagel's lambda, AIC = Akaike Information
790 Criterion, w_i = Akaike weight. Akaike weights were calculated from AIC values of all models for each trait. Models are ranked by AIC and only
791 models with $\Delta AIC < 2$ are presented. See Appendix S2 in Supplementary Material for a complete list of analyzed models.

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		Predictors in the model													F		
Physiological trait	Region	TMAX	tmax	TMIN	tmin	TMEAN	tmean	SEASONAL RANGE	seasonal range	daily range	λ	AIC	ΔAIC	w_i	d.f	Value	P
CTmax	Global		0.487			1.111					0.73	113.1	0	0.16	2,35	18.56	<0.01
			0.659			0.945			-0.248		0.7	114.9	2	0.06	3,34	12.29	<0.01
	Temperate							0.950		0.724	0.85	53.1	0	0.07	2,14	4.50	0.03
								1.053	0.753		0.97	53.6	0.5	0.05	2,14	7.28	<0.01
								1.056	-0.910	1.397	0.79	53.7	0.6	0.05	3,13	3.14	0.06
	Tropical		0.653								0	54.2	0	0.07	1,19	13.07	<0.01
									0.905		0	54.7	0.5	0.06	1,19	12.34	<0.01
								2.159	1.395	-0.383	0	55.0	0.8	0.05	3,17	5.47	<0.01
CTmin	Global			1.628	1.867						0	128.2	0	0.1	2,35	153.80	<0.01
				1.304	1.322		0.881				0	128.3	0.1	0.1	3,34	105.10	<0.01
				1.453			1.746		-0.463		0	128.8	0.6	0.08	3,34	103.70	<0.01
	Temperate				3.204						0	55.2	0	0.07	1,15	7.69	0.01
					2.120		1.008				0	55.6	0.4	0.06	2,14	4.63	0.03
					3.228	0.931					0	55.7	0.5	0.06	2,14	4.54	0.03
	Tropical						3.334	-5.638	-0.738		0.77	69.0	0	0.06	3,17	4.02	0.02
				4.501			1.381				0.76	69.2	0.2	0.06	2,18	4.78	0.02
						7.894					0.7	69.8	0.8	0.04	1,19	6.43	0.02
Tolerance Range	Global							1.155	1.205		0	158.1	0	0.48	2,35	17.85	<0.01
								1.024	1.566	-0.389	0	159.4	1.4	0.24	3,34	11.93	<0.01
	Temperate								0.622		0.22	65	0	0.27	1,15	0.92	0.35
	Tropical							5.166	0.749		0.43	86.5	0	0.28	2,18	2.58	0.1

798 **FIGURES**

799 **Figure 1.** Upper and lower critical thermal limits (CTmax an CTmin) for tropical and
800 temperate tadpole assemblages. Dashed lines indicate the average CTmax / CTmin and
801 95% CI, respectively, for the overall geographical area. Species codes, ordered
802 alphabetically within region, see Table S1.

803 **Figure 2.** (a) CTmax and CTmin variation in tadpoles with the absolute latitude (ºN or
804 ºS) of the centroid´s distribution. The change is steeper for CTmin than CTmax
805 (ANCOVA $p<0.01$). (b) Phylogenetic generalized least squares for tolerance range and
806 absolute latitude of the poleward limit of distribution.

807 **Figure 3.** Phylogenetic generalized least squares for thermal range and the log-
808 transformed of the area distribution (range size) (a) and latitudinal range (b) of the
809 analyzed tadpoles.

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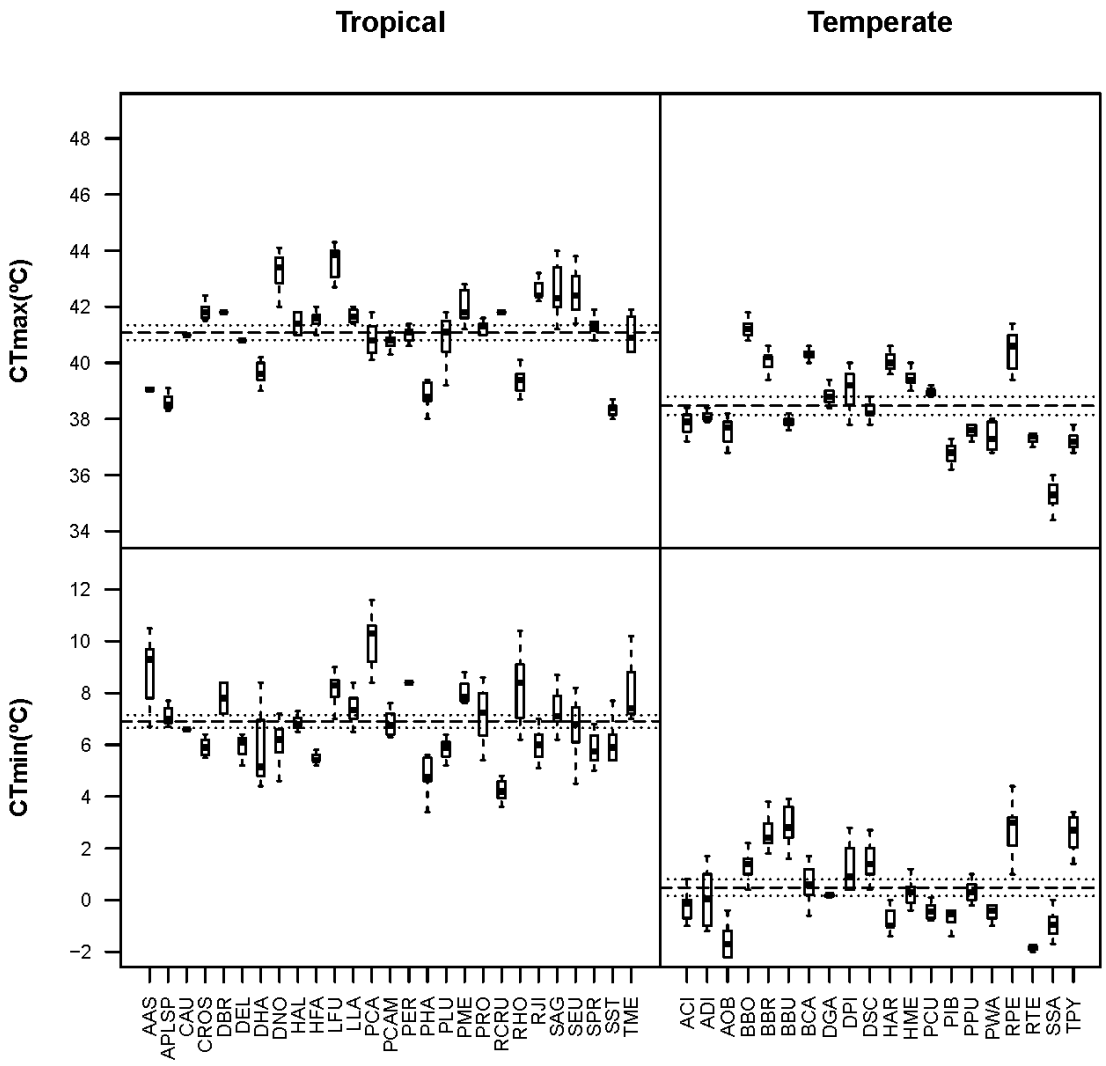
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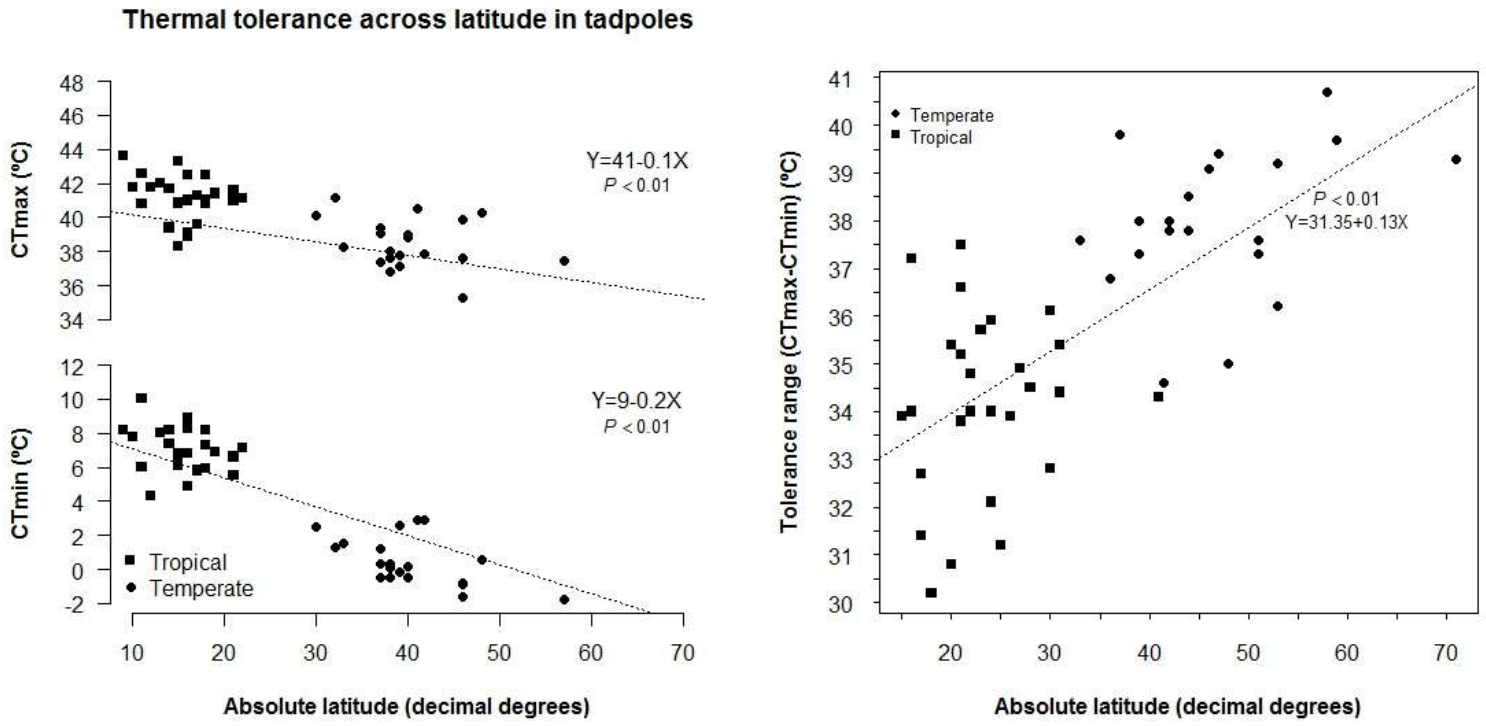
824 **Figure 1.**



825

826 **Figure 2**.

827 a b

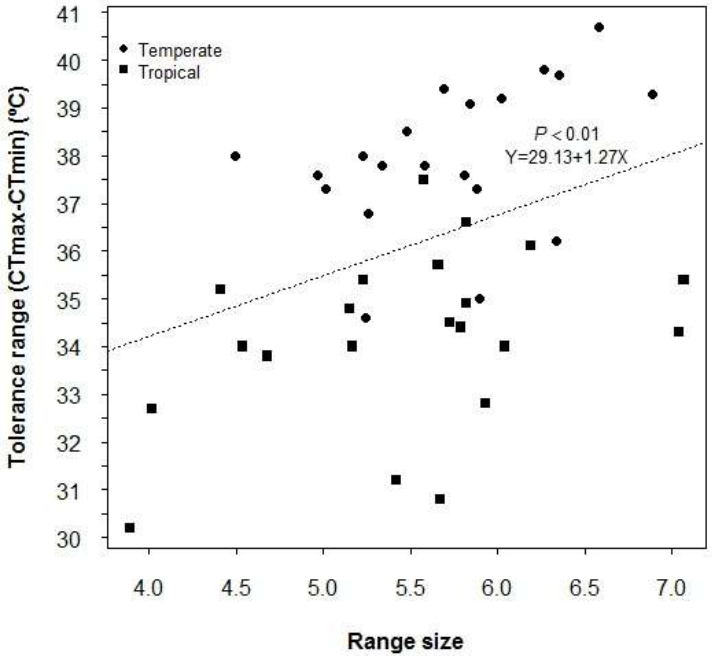


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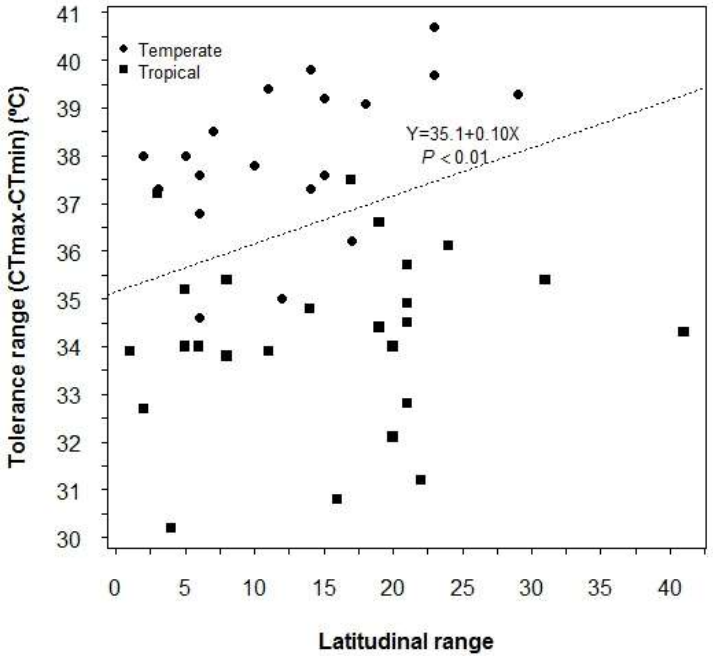
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Figure 3.

a


Temperate
Tropical
P < 0.01
Y=29.13+1.27X
Tolerance range (CTmax-CTmin) (ºC)
Range size
41
40
39
38
37
36
35
34
33
32
31
30
4.0
4.5
5.0
5.5
6.0
6.5
7.0

b


Temperate
Tropical
Y=35.1+0.10X
P < 0.01
Tolerance range (CTmax-CTmin) (ºC)
Latitudinal range
41
40
39
38
37
36
35
34
33
32
31
30
0
5
10
15
20
25
30
35
40